Astro Night Planner 1.4.2

Practical manual for astrophotography planning with weather, Moon, object selection, equipment, horizon, framing and local surveys

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Contents

Overview

Astro Night Planner combines visibility, twilight, Moon, weather, location, horizon, equipment and Aladin framing in one planning interface. Profiles, equipment, locations and settings are stored locally in the browser. Create an external backup after important changes.

Version 1.4.2 adds integrated survey downloads, operating-system folder dialogs and a clear bulk switch for local sources. The Windows solution with automatically opened browser interface, tray and autostart and the native Linux/macOS edition remain available.

Planning night, date buttons and Moon

Date buttons summarize the selected night. Weather and Moon values refer to the selected planning window, not automatically to the entire night.

Indicator	Meaning
↑	Moonrise, if it occurs in the relevant night window.
▲	Maximum Moon altitude **within the selected planning window**.
↓	Moonset. If it is outside the planning window, the planning data says so explicitly.

The planning data may also show the Moon culmination of the night. If it is before or after the planning window, the app adds a corresponding note.

Equipment and Aladin framing

Telescope, reducer/corrector/Barlow and camera determine effective focal length, field of view, pixel scale and the camera frame. In the Aladin view, combinations can be tried temporarily. If a combination is not saved as a setup, it still remains active and can be saved as a new setup.

Example: 700 mm focal length with a 0.80x reducer gives 560 mm effective focal length. With the same camera the field of view becomes larger.

Local surveys and Local Survey Server

Local surveys are local copies of HiPS sky maps served through a local HTTP server. The app and Aladin use URLs such as `http://127.0.0.1:8765/nsns/ohs8/properties`.

Common principle

```
Local survey base URL: http://127.0.0.1:8765/
Relative path:         nsns/ohs8/
```

The shared root can be `D:\AstroSurveys` on Windows, `/home/user/AstroSurveys` on Linux or `/Users/user/AstroSurveys` on macOS.

Windows

ANP Local Survey Server 1.1 (Build 1.1.4) automatically opens its browser interface on double-click. Tray, autostart and background operation remain available.

Tray action	Effect
Open interface	Reopens the browser interface.
Start server	Starts the survey file server.
Stop server	Stops only the file server; tray and configuration remain active.
Exit	Stops all components and releases the EXE.

The Windows folder dialog is native and requires neither PowerShell nor a PATH entry. Existing configuration from the previous server version is detected and migrated.

Linux and macOS

The native package includes binaries for Linux AMD64/ARM64 and macOS Intel/Apple Silicon. Browser interface and download logic match the Windows edition. Linux folder selection uses `zenity` or `kdialog`; macOS uses the system Finder dialog. Autostart is configured with `systemd --user` on Linux and a LaunchAgent on macOS.

Survey downloads

The download manager includes all 15 surveys built into the app with verified default service URLs: DSS2 color/red/blue, PanSTARRS, VTSS, Finkbeiner, AllWISE, 2MASS, IRIS and six NSNS surveys. Custom HiPS sources are also supported.

Before downloading, `properties` and `Moc.fits` are validated. Required tiles are derived from the MOC coverage map; no directory listing is needed. Missing target folders are created automatically. `.part` files allow resume and complete existing files are skipped.

Save source and target path: Both values are stored per survey. The resolved local target path is shown. Absolute Windows paths may use backslashes; relative HiPS paths use forward slashes and are normalized internally.

A complete table of verified default service URLs and target paths is available in the [survey download HTML guide].

Folder selection in Planner

The Planner needs a path relative to the server root. It therefore does not open a general file dialog; it queries the running server and presents all detected HiPS folders containing `properties` in its own dialog.

Global and individual preference

Prefer local sources for all surveys sets or clears **Prefer local source** in every survey row. Individual surveys can then be changed separately. Online fallback is used only when enabled.

Python fallback

The Python package remains available as a transparent expert and fallback option.

Weather, cloud model and external sources

Weather, cloud model, Meteoblue, Windy, Ventusky and other external reference sources support the decision whether a night is suitable for astrophotography. External services require an internet connection.

Backup, PWA and troubleshooting

The app uses local browser data. Create regular full backups. Local survey requests to `127.0.0.1`, `localhost` and private LAN addresses are not intercepted by the PWA cache but passed directly to the local server.

- If local surveys do not work, first check `http://127.0.0.1:8765/nsns/ohs8/properties` in the browser.
- Before replacing the Windows server package, use **Exit** in the tray menu.
- If Linux/macOS should start in the background, configure systemd user service or LaunchAgent.

Glossary

Term	Explanation
HiPS	Hierarchical Progressive Survey, a tiled sky-map format.
properties	Control file of a HiPS survey with tile format and resolution.
Norder	Folder structure of HiPS tiles by resolution order.
Fallback	Switching to the online source if the local source is not reachable and fallback is enabled.

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